In the preceding example, finding all devices in a given home would be easy, but finding what home a particular device is in would be significantly harder.

4.3.5 Key-value structures

Key-value structures provide the simplest data model, storing data as pairs of unique identifiers and their associated values. This approach optimizes for high-performance lookups and scales horizontally across distributed systems.

While key-value structures sacrifice query flexibility, they excel at use cases with predictable access patterns and high-performance requirements. Caching frequently accessed reference data or storing session information are common key-value applications.

4.4 Data ingestion

While accessing data in place offers simplicity and reduces duplication, digital twins often require dedicated data ingestion to meet their operational and analytical requirements. The decision to import rather than query external systems may be driven by the need to transform and enrich source data, run advanced analytics that might overwhelm transactional systems, or maintain operational independence from source systems. Lets look at some of the ways in which a digital twin can ingest data.

4.4.1 Batch data ingestion

Batch ingestion processes data in discrete chunks at scheduled intervals, hourly, daily, or on-demand. Batch ingestion is popular for slow moving enterprise data and offers advantages in reliability, cost efficiency, and processing complex transformations. It is an important mechanism for a digital twin because it efficiently handles large volumes of historical, contextual, and less time-sensitive data, providing a comprehensive and stable foundation for analysis, modeling, and long-term insights. This type of data is used by digital twins to understand past behavior in order to predict future outcomes.

TRY IT OUT: INGEST MILLIONS OF TAXI TRIPS TO MODEL CITY TRANSPORTATION

Imagine that you are building a city scale digital twin of New York. There are over 13,000 iconic yellow cabs in the city making millions of trips per year. Understanding these taxi movements in the real world enables your digital twin to gain insight into the functioning of the city’s transportation network, enabling congestion management, infrastructure planning, and simulation and testing of new traffic management strategies. The New York City Taxi and Limousine Commission provides open taxi trip record data published monthly at https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page. We can ingest this data in monthly batches, cleansing it on ingestion, and store it in a relational database within our